PATRICK YIN

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EDUCATION

University of Washington 2023–
Ph.D. Computer Science & Engineering
Awards: NSF Graduate Research Fellow, ∼ 10% selection rate

University of California, Berkeley 2019–2023
B.A. Computer Science, GPA: 4.0/4.0
Awards: Regents' and Chancellor's Scholar, top <2% incoming class; National Merit Scholar

EXPERIENCE

Research Intern, NVIDIA — *Advised by Karl Van Wyk, Ankur Handa, Nathan Ratliff* 2026–
Working on reinforcement learning and sim2real transfer for dexterous hands.

Graduate Researcher, UW Robotics — *Advised by Abhishek Gupta* 2023–
Working on reinforcement learning and sim2real transfer for dexterous robot manipulation.

Student Researcher, Microsoft Research — *Advised by Andrey Kolobov* 2025–2026
Research Intern, Microsoft Research 2025
Worked on reinforcement learning and sim2real transfer for dexterous robot manipulation.

Undergrad Researcher, Berkeley AI Research — *Advised by Sergey Levine, Kuan Fang, Ashvin Nair* 2020–2023
Published 4 papers on offline goal-conditioned reinforcement learning and finetuning for real-world robotic control.

Machine Learning Engineer Intern, Ambi Robotics — *Advised by Yi Li* 2022
Spearheaded training deep learning models on real-world production data and training 3D neural networks.

Software Engineer Intern, UiPath 2021
Pushed over 30 Git commits to production on Insights team working with Snowflake/SQL, Kubernetes, and Docker.

SELECTED PUBLICATIONS

A Balanced Data Diet: Addressing the Bottleneck in Mega-Scale RL for Robot Control
O Zhang*, M Castro*, **Patrick Yin***, Ignacio Dagnino, A Gupta, R Scalise, B Boots
Conference on Robot Learning (CoRL), 2026

OmniReset: Emergent Dexterity Via Diverse Resets and Large-Scale Reinforcement Learning
Patrick Yin*, T Westenbroek*, O Zhang, J Tran, I Dagnino, E Shilamkar, N Tiapo, S Bagaria, X Liu, G Mullins, A Kolobov, A Gupta
International Conference on Learning Representations (ICLR), 2026

OTHER PUBLICATIONS

Visual Sim-to-Real Learning for Robotic Insertion under Geometric Variations: Rebar Installation
T Sun*, B Han*, **Patrick Yin**, R Xu, H He, A Gupta, S Rusinkiewicz, Y Shao
In Submission

Simulation Distillation: Pretraining World Models in Simulation for Rapid Real-World Adaptation
J Levy*, T Westenbroek*, K Huang, F Palafox, **Patrick Yin**, S Omidshafiei, D Kim, A Gupta, D Fridovich-Keil
Robotics: Science and Systems (RSS), 2026

SGFT: Rapidly Adapting Policies to the Real-World via Simulation-Guided Fine-Tuning
Patrick Yin*, T Westenbroek*, S Bagaria, K Huang, C Cheng, A Kolobov, A Gupta
International Conference on Learning Representations (ICLR), 2025

DROID: A Large-Scale In-the-Wild Robot Manipulation Dataset
A Khazatsky*, K Pertsch*, ..., **Patrick Yin**, et al.
Robotics: Science and Systems (RSS), 2024

ASID: Active Exploration for System Identification and Reconstruction in Robotic Manipulation
M Memmel, C Zhu, A Wagenmaker, **Patrick Yin**, D Fox, A Gupta
International Conference on Learning Representations (ICLR), 2024 (Oral Presentation)

Stabilizing Contrastive RL: Techniques for Robotic Goal Reaching from Offline Data
C Zheng, B Eysenbach, H Walke, **Patrick Yin**, K Fang, R Salakhutdinov, S Levine
International Conference on Learning Representations (ICLR), 2024 (Spotlight Talk)

Open X-Embodiment: Robotic Learning Datasets and RT-X Models
Open X-Embodiment Collaboration, ..., **Patrick Yin**, et al.
IEEE International Conference on Robotics and Automation (ICRA), 2024 (Best Paper)

Generalization with Lossy Affordances: Leveraging Broad Offline Data for Learning Visuomotor Tasks
K Fang, **Patrick Yin**, A Nair, H Walke, G Yan, S Levine
Conference on Robot Learning (CoRL), 2022 (Oral Presentation)

Planning to Practice: Efficient Online Fine-Tuning by Composing Goals in Latent Space
K Fang*, **Patrick Yin***, A Nair, S Levine
IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), 2022

Bisimulation Makes Analogies in Goal-Conditioned Reinforcement Learning
P Hansen-Estruch, A Zhang, A Nair, **Patrick Yin**, S Levine
International Conference on Machine Learning (ICML), 2022

AWARDS

NSF Graduate Research Fellowship. $\sim$ 10% selection rate. 2025
Highest Distinction in General Scholarship, L&S CS. Top $\sim$ 3% of graduating class at Cal. 2023
Regents' and Chancellor's Scholar. Top $<$ 2% of incoming class at Cal. 2019
National Merit Scholar. Top 0.02% of high school seniors in US. 2019

MENTORSHIP

Yanda Bao — BS @ UW
Joshua Tran — BS @ UW
Ignacio Dagnino — Student @ Skyline High School $\rightarrow$ Researcher @ UW
Simran Bagaria — BS/MS @ UW $\rightarrow$ Research Engineer @ Microsoft Research $\rightarrow$ Research Engineer @ Eka Robotics

PROFESSIONAL SERVICE

Teaching Assistant
CSE 542: Reinforcement Learning 2024

Reviewer
IEEE Transactions on Robot Learning (T-RL) 2026
IEEE International Conference on Robotics and Automation (ICRA) 2025,2026
Computer Vision and Pattern Recognition Conference (CVPR) 2024,2025
Conference on Robot Learning (CoRL) 2024

UW CSE PhD Admissions Reviewer 2024

OUTREACH AND SERVICE

UW Robotics Lab Outreach Coordinator 2023–2024
UAW 4121 Cohort Liaison 2023–2024
UW U-PASS Student Advisory Board Member 2023–2024
Pre-Application Mentorship Service (PAMS) Mentor 2023